Computing the weighted sum of values: σV

remember $V = \begin{bmatrix} 1, 0 \\ 0, 2 \\ 3, 1 \end{bmatrix}$. ~~we~~ compute $O = \sigma V$.

- Toke 1

$\sigma_1 = [0.401, 0.401, 0.198]$ $O_1 = 0.401 v_1 + 0.401 v_2 + 0.198 v_3$

*first component

$O_{1,1} = 0.401(1) + 0.401(0) + 0.198(3) \approx 0.401 + 0 + 0.594 \approx 0.995$

*Second component

$O_{1,2} = 0.401(0) + 0.401(2) + 0.198(1) = 0 + 0.802 + 0.198 \approx 1.000$

So $O_1 \approx [0.995, 1.000] \approx [1.0, 1.0]$

- Token 2

$\sigma_2 = [0.198, 0.401, 0.401] \rightarrow O_2 = 0.198 v_1 + 0.401 v_2 + 0.401 v_3$

*First component

$O_{2,1} = 0.198(1) + 0.401(0) + 0.198(3) = 0.198 + 0 + 1.203 \approx 1.401$

*Second component

$O_{2,2} = 0.198(0) + 0.401(2) + 0.401(1) = 0 + 0.802 + 0.401 = 1.203$

So $O_2 \approx [1.401, 1.203]$

- Token 3

$\sigma_3 = [0.248, 0.503, 0.248]$ $O_3 = 0.248 v_1 + 0.503 v_2 + 0.248 v_3$

*First Componente

$O_{3,1} = 0.248(1) + 0.503(0) + 0.248(3) \approx 0.248 + 0 + 0.744 = 0.992$

*Second Component

$O_{3,1} = 0.248(0) + 0.503(2) + 0.248(1) = 0 + 1.006 + 0.284 \approx 1.254$

So $O_3 = [0.992, 1.254]$

Final Attention output $O = \sigma V \approx \begin{bmatrix} 0.995 & 1.000 \\ 1.401 & 1.203 \\ 0.992 & 1.254 \end{bmatrix}$

Each row is a contextualized representation of the corresponding token, formed as a weighted mixture of all value vectors.

QK^T: raw similarity btw "what I'm looking for" (Query) and what each token is

Divide by $\sqrt{d_k}$: keep scores in a reasonable range → no saturation for Softmax

Softmax row-wise: Scores into attention distribution → each row is how much token i attends to each token j

Multiply by V → Token's new representation is the expected value of V under that attention distribution